

Ex No 10:

//Step 2: Create a database use library

//switched to db library

//Step 3: Insert data db.authors.insertMany([

{ "AuthorID": 1, "FirstName": "George", "LastName": "Orwell" },

{ "AuthorID": 2, "FirstName": "Aldous", "LastName": "Huxley" },

{ "AuthorID": 3, "FirstName": "J.K.", "LastName": "Rowling" }

])

//Output: { "acknowledged": true, "insertedIds": [1, 2, 3] }

db.books.insertMany([

{ "BookID": 1, "Title": "1984", "Genre": "Dystopian", "PublicationYear": 1949, "Authors": [1] },

{ "BookID": 2, "Title": "Brave New World", "Genre": "Dystopian", "PublicationYear": 1932, "Authors":

[2] },

{ "BookID": 3, "Title": "Harry Potter and the Sorcerer's Stone", "Genre": "Fantasy", "PublicationYear":

1997, "Authors": [3] }

])

//Output: { "acknowledged": true, "insertedIds": [1, 2, 3] }

db.borrowers.insertMany([

{ "BorrowerID": 1, "FirstName": "John", "LastName": "Doe", "MembershipDate": new Date("2023-01-01") },

{ "BorrowerID": 2, "FirstName": "Jane", "LastName": "Smith", "MembershipDate": new Date("2023-02-15") }

])

//Output: { "acknowledged": true, "insertedIds": [1, 2] }

db.borrowedBooks.insertMany([

{ "BorrowerID": 1, "BookID": 1, "BorrowedDate": new Date("2023-03-01"), "ReturnDate": new Date("2023-03-15") },

{ "BorrowerID": 2, "BookID": 3, "BorrowedDate": new Date("2023-03-05"), "ReturnDate": new Date("2023-03-20") }

])

// Output: { "acknowledged": true, "insertedIds": [1, 2] }

// Step 4: Update data

```
db.authors.updateOne({ "AuthorID": 1 }, { $set: { "LastName": "Smith" } })
```

//Output: { "acknowledged": true, "matchedCount": 1, "modifiedCount": 1 }

//Step 5: Query data db.authors.find().pretty()

//Output:

```
//{ "_id": ObjectId("..."), "AuthorID": 1, "FirstName": "George", "LastName": "Smith" }
```

```
//{ "_id": ObjectId("..."), "AuthorID": 2, "FirstName": "Aldous", "LastName": "Huxley" }
```

```
//{ "_id": ObjectId("..."), "AuthorID": 3, "FirstName": "J.K.", "LastName": "Rowling" }
```

//Step 6: Delete data db.authors.deleteOne({ "AuthorID": 1 })

//Output: { "acknowledged": true, "deletedCount": 1 }

Output:

MongoDB> use library switched to db library

MongoDB> db.authors.insertMany([...])

```
{
  acknowledged: true, insertedIds: [ 1, 2, 3 ]
}
```

MongoDB> db.books.insertMany([...])

```
{
  acknowledged: true, insertedIds: [ 1, 2, 3 ]
}
```

MongoDB> db.borrowers.insertMany([...])

```
{
  acknowledged: true, insertedIds: [ 1, 2 ]
}
```

MongoDB> db.borrowedBooks.insertMany([...])

```
{
  acknowledged: true, insertedIds: [ 1, 2 ]
}
```

MongoDB> db.authors.updateOne(

... { AuthorID: 1 },

... { \$set: { LastName: "Smith" } })

```
{
```

acknowledged: true, matchedCount: 1,

modifiedCount: 1

```
}
```

MongoDB> db.authors.find().pretty()

```
{
```

"_id" : ObjectId("..."), "AuthorID" : 1, "FirstName" : "George", "LastName" : "Smith" }

```
{
```

"_id" : ObjectId("..."), "AuthorID" : 2, "FirstName" : "Aldous", "LastName" : "Huxley"

```
}
```

```
{
```

```
"_id" : ObjectId("..."), "AuthorID" : 3, "FirstName" : "J.K.", "LastName" : "Rowling"  
}
```

```
MongoDB> db.authors.deleteOne({ AuthorID: 1 })
```

```
{  
  acknowledged: true, deletedCount: 1  
}
```